

Digital product passports on interoperable cloud data platforms to help smallholder agri-food cooperatives meet EU sustainability compliance and access premium export markets.

Wholesale × Product traceability and digital product passport × Cloud data platform · OS033 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
67 is the world moving	37 can we play, can we win	HIGH 0 named in the evidence	€1.1bn modelled evidence · per year	NEXT regulation adopted or propose...	L2 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 0.0 published reference(s) in Wholesale, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

Digital product passports on interoperable cloud data platforms can help smallholder agri-food cooperatives meet EU sustainability compliance and access premium export markets. This opportunity exists now because the EU's Ecodesign for Sustainable Products Regulation is making DPPs mandatory for a wide range of product categories, and the EU Commission has launched the Digital Product Passport Registry, creating an urgent need for interoperable data platforms.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€1.8bn €521m – €6.5bn	€1.1bn €310m – €3.8bn	€2.7m €774k – €9.6m	modelled

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

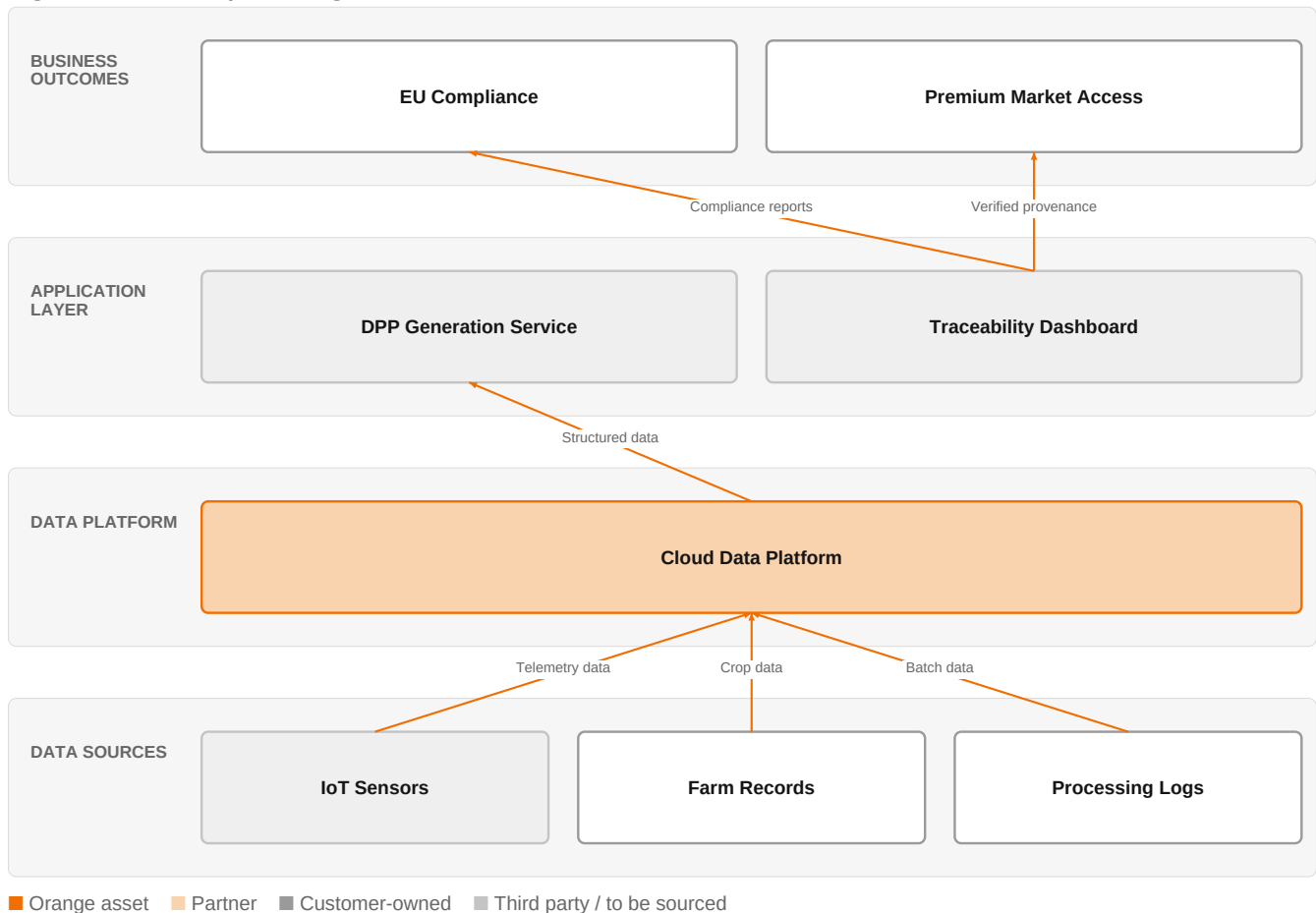
Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Wholesale (EU27_2020)	148,208 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Cloud data platform	31.4 % of enterprises (10+ employees)	observed	Eurostat · isoc_cicce_usen2 · E_CC_PDB	2025
Annual value of one engagement	€320k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	18,053 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract value is a configured assumption, not observed. Only 0 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 2 declared geographies are outside the European reference data (LK); the estimate covers the European footprint only.

The solution, in one picture

Digital Product Passport for Agri-Food



This diagram shows how Orange integrates IoT data from farms and processing into a cloud platform to generate digital product passports that enable EU compliance and premium market access.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU's Ecodesign for Sustainable Products Regulation (ESPR) mandates Digital Product Passports for a wide range of product categories, creating an urgent need for machine-readable product lifecycle data. The EU Commission has launched the Digital Product Passport Registry, raising new data demands for businesses. Smallholder agri-food supply chains face persistent barriers in accessing high-value and sustainable markets, including regulatory compliance challenges and limited traceability, and DPPs are seen as a pathway to overcome these barriers.

Sources: SIG-AA85D6CD0C67 openalex.org 2026-02-27; SIG-BA196FF05244 Springer Science and B 2026-02-03; SIG-C4998B3557D1 INSIGHT EU MONITORING 2026-07-22; SIG-A3D483674C44 Packaging Insights 2026-08-03; SIG-25966E62DFAB openalex.org 2026-02-04

Who buys, and why now

The buyer is typically the chief sustainability officer or head of export at a smallholder agri-food cooperative, or the IT director at a cooperative union. The pain is operational: they cannot demonstrate compliance with EU sustainability regulations, and they are locked out of premium export markets because buyers demand traceability and transparency they cannot provide. The trigger is an export order that requires DPP compliance, or a regulatory deadline that makes non-compliance a business risk.

Why it is hot now — every claim bound to a dated source

- Digital product passports can help smallholder agri-food supply chains overcome barriers to accessing high-value and sustainable markets. [SIG-25966E62DFAB]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a cloud data platform (L3) using its AI/Data/Cloud experts (SUP) and IoT experts (SUP) to build a digital product passport solution. The engagement would start with a discovery phase to map the cooperative's supply chain data, then design and deploy a cloud-based platform that ingests IoT data from farms and processing sites, and generates DPPs that comply with EU standards. Orange Group researchers (SUP) would contribute to ensuring interoperability with the EU registry. The platform would be hosted on a hyperscaler (Microsoft, AWS, or Google Cloud) as a partner, with Orange providing the integration and certification (SOC 2 Type II) to assure trust.

Why Orange can win

Orange can win because it combines deep expertise in AI, data, and cloud (SUP) with IoT capabilities (SUP) and research backing (Orange Group researchers), and it can deliver on a trusted, certified infrastructure (SOC 2 Type II). The hyperscalers are both partners and competitors, but Orange's role as an integrator and its focus on trust and sovereignty give it an edge in a market where data integrity and compliance are paramount. However, the assets are thin: Orange does not have a pre-built DPP solution, so the win depends on the strength of its consulting and integration capabilities.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Microsoft	partner	L2	partner provides the required technology	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
Cloud data platform	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
SOC 2 Type II	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

HIGH competitive intensity (score 8.6; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitors include hyperscalers AWS, Google Cloud, and Microsoft, who sell cloud data platforms and can offer DPP solutions as part of their broader cloud offerings. Global systems integrators like Accenture, Capgemini, DXC Technology, Infosys, and Sopra Steria also sell cloud data platforms and have the consulting muscle to implement DPPs. The customer will compare Orange against these players, who are strong in their respective areas: hyperscalers in platform scale and ecosystem, and integrators in end-to-end delivery. Orange's differentiation must come from its trusted, sovereign positioning and its ability to orchestrate across partners.

Sources: SIG-AA85D6CD0C67 openalex.org 2026-02-27; SIG-BA196FF05244 Springer Science and B 2026-02-03; SIG-C4998B3557D1 INSIGHT EU MONITORING 2026-07-22

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Cloud data platform; competes in OX: Smart Industries — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Cloud data platform — also an Orange partner	structural	—
Microsoft	Hyperscaler	sells Cloud data platform — also an Orange partner	structural	—
Accenture	Global systems integrator	sells Cloud data platform; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	sells Cloud data platform; competes in OX: Smart Industries	structural	—
DXC Technology	Global systems integrator	sells Cloud data platform	structural	—
Infosys	Global systems integrator	sells Cloud data platform	structural	—

Sopra Steria	Global systems integrator	sells Cloud data platform	structural	—
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evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 Do you currently export to the EU, and have you been asked by any buyer to provide product traceability data or a digital product passport?
- 2 What is your current process for tracking products from farm to export, and where are the biggest gaps in data collection?
- 3 Have you started preparing for the EU's Ecodesign for Sustainable Products Regulation, and if so, what is your biggest challenge?
- 4 Do you have any existing IoT or data collection systems on your farms or in your processing facilities?
- 5 Who in your organization would be responsible for ensuring compliance with EU sustainability regulations, and how are they currently managing it?
- 6 Are you working with any technology partners today for traceability or supply chain management, and what has been your experience?

Objections you will meet

They will say	You can answer
We are a small cooperative; we don't have the budget or IT skills for a digital product passport system.	That's a fair concern. Many smallholders face the same challenge. Our approach is to start small, using a cloud platform that scales with your needs, and we provide the expertise to manage it for you. The investment can be offset by the premium prices you can command in export markets once you have DPPs in place.
The EU regulations are still evolving; we don't want to invest in something that might change.	You're right that standards are evolving. That's why we design our solution to be flexible and interoperable, so it can adapt to new requirements. We also stay on top of regulatory changes, so you don't have to.
We already work with a hyperscaler for our IT; why do we need Orange?	That's a good point. Hyperscalers provide great platforms, but they don't offer the end-to-end integration and domain expertise that we do. We can help you design a DPP solution that works across your entire supply chain, and we bring certified security and trust that is critical for compliance.

Risks and unknowns

The main risk is that DPP standards are still evolving, and the EU registry is new, so the solution may need to adapt to changing requirements. There is also uncertainty about the willingness of smallholder cooperatives to invest in digital infrastructure, given their limited resources. Additionally, the competitive intensity is high, and Orange's assets are not yet proven in this specific domain, so the outcome depends on execution.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief that outlines how Orange's AI / Data / Cloud experts and IoT experts, leveraging a cloud data platform and SOC 2 Type II certification, can deliver an interoperable digital product passport solution for smallholder agri-food cooperatives, differentiating our approach through our partner ecosystem.
Sales	In a conversation with a wholesale agri-food cooperative's sustainability or export lead, say: 'Orange's cloud data platform expertise, backed by SOC 2 Type II certification and partnerships with Microsoft, AWS, and Google Cloud, can help you meet EU sustainability compliance and unlock premium export markets through digital product passports.'
Strategist / Innovator	Commission a deep-dive brief on how Orange's AI / Data / Cloud experts and IoT experts could support interoperable cloud data platforms for digital product passports in smallholder agri-food cooperatives, focusing on EU sustainability compliance and premium export market access.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-E4163170FA87	2026-06-17	Systems	1	Unlocking Digital Product Passport Integration: Multidimensional Hurdles in Supply Chains
SIG-36FFB2435F8A	2026-04-20	openalex.org	1	A Comprehensive Review of Digital Product Passports in Sustainable Manufacturing: Current State...
SIG-6B43CC88BC8E	2026-03-14	openalex.org	1	Integrating knowledge management and digital product passports to foster sustainable and collab...
SIG-AA85D6CD0C67	2026-02-27	openalex.org	1	Digital product passports
SIG-25966E62DFAB	2026-02-04	openalex.org	1	Transforming smallholder agri-food supply chains through digital product passports: A pathway t...
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-EC1CEFFD337A	2025-10-15	openalex.org	1	Digital Product Passport in support of Data-Driven End-of-Use Strategies—a systems design persp...
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-F0C3FBE1B02D	2026-08-18	The Chartered Institute o...	2	Government looks for businesses to input on digital product passports - The Chartered Institute...
SIG-A3D483674C44	2026-08-03	Packaging Insights	2	EU's Digital Product Passport Registry raises new data demands - Packaging Insights
SIG-C4998B3557D1	2026-07-22	INSIGHT EU MONITORING	2	EU Commission launches Digital Product Passport Registry - INSIGHT EU MONITORING
SIG-370D30A66C8C	2026-07-21	EU NEIGHBOURS east	2	EU's Digital Product Passport Registry goes live - EU NEIGHBOURS east
SIG-4FE0689B82DF	2026-04-30	Packaging Digest	2	EU Digital Product Passport Impact on US Supply Chains - Packaging Digest

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (3 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:13+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:42:24+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.